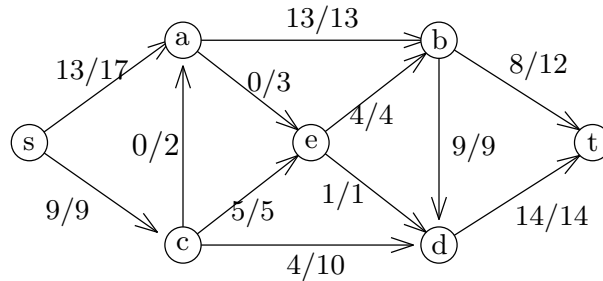


CS 473 (Spring 2025)

Midterm 2 Practice Problems

Short Questions

1. Run *one iteration* of the Ford–Fulkerson algorithm for the maximum flow problem on the following example, starting with the initial flow shown. Here, the x/y label on an edge indicates that the edge has capacity y and initial flow value x . Show the residual graph G_f , and the augmenting path chosen (if there are choices, pick any you prefer), and the new flow. Is the new flow after one iteration optimal, and why?



2. True or False: If we are unlucky, it is possible that Sharir and Welzl's randomized incremental algorithm for the smallest enclosing circle problem may take $\Omega(n^2)$ time. Briefly justify your answer.
3. What is Chebyshev's inequality? Explain how it can be derived from Markov's inequality.
4. What goes wrong when adapting the Hungarian method to find maximum matchings in general, non-bipartite graphs? Briefly explain.

Randomized Algorithms

1. A *palindrome* is a string that reads the same forward and backward. Describe a Las Vegas algorithm that solves the following problem in $O(n)$ expected time given a string $a_1a_2 \cdots a_n \in \{0, 1\}^*$ of length n , find the *longest palindromic prefix*, i.e., the largest i such that $a_1a_2 \cdots a_i$ is a palindrome.

Example (over the English alphabet): for the input string MADAMIMADAMHELLOADAM, the answer is MADAMIMADAM.

2. We are given two $n \times n$ matrices A and B , where all entries are in $\{0, 1\}$. We want to decide whether $AB = BA$. With Strassen's algorithm, this problem can be solved in $O(n^{2.81})$ time. Using randomization, describe a Monte Carlo algorithm that solves this problem in $O(n^2)$ time, with error probability at most 0.1.

(Hint: pick a random $x \in \{0, \dots, p-1\}$ for a fixed prime p , define vector $v = (1, x, x^2, \dots, x^{n-1})^T$, and compute ABv and BAv .)

3. Given an undirected graph $G = (V, E)$ with n vertices, a *3-coloring* is a mapping $\phi : V \rightarrow \{1, 2, 3\}$ such that for every $uv \in E$, we have $\phi(u) \neq \phi(v)$. We can find a 3-coloring by brute force in time $O(3^n)$ up to a polynomial factor. In this question, you will explore a faster randomized approach.
 - (a) Consider a variant of the problem, where we are given an extra mapping $f : V \rightarrow \{1, 2, 3\}$ and we want to find an *f-avoiding 3-coloring*, defined as a 3-coloring ϕ such that $\phi(v) \neq f(v)$ for every $v \in V$. Show that this problem reduces to 2SAT and can thus be solved in polynomial time. (Hint: for each vertex, create 3 variables; then add clauses of length 2. . .)
 - (b) Define f by setting each $f(v)$ independently at random from $\{1, 2, 3\}$. For a fixed 3-coloring ϕ , calculate the probability that ϕ is *f-avoiding*.
 - (c) Using (a) and (b), give a Monte Carlo algorithm to find a 3-coloring in $O(c^n)$ time for some constant c better than 3, with error probability less than 0.01. (Note: the inequality $1 - x \leq e^{-x}$ might be useful.)

Matching, Flows, and Cuts

1. Given an *undirected* graph $G = (V, E)$ with two special vertices $s, t \in V$, we want to find the smallest subset of vertices $S \subseteq V - \{s, t\}$ whose removal disconnects t from s . Show how this problem can be solved efficiently by using a max flow algorithm. Prove correctness. (Hint: define a new directed graph G' with twice the number of vertices. . .)
2. (Tricky?) Given a directed graph $G = (V, E)$ with source s and sink t , where each edge e has a capacity $c(e)$, we want to find a flow f with maximum value, such that $1 \leq f(e) \leq c(e)$ (instead of the usual constraint $0 \leq f(e) \leq c(e)$ in the standard version of the max flow problem). Show how this problem can be reduced to the standard version of the max flow problem.